

Introduction to Sensors & Programming

HOW ROBOTS “SEE,” “FEEL,” AND MAKE DECISIONS

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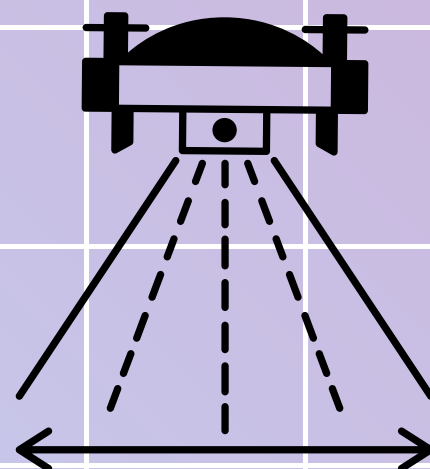
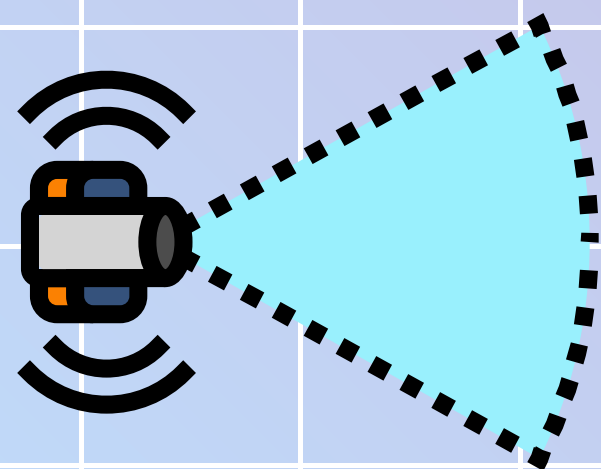
What Is a Sensor?

- DEFINITION

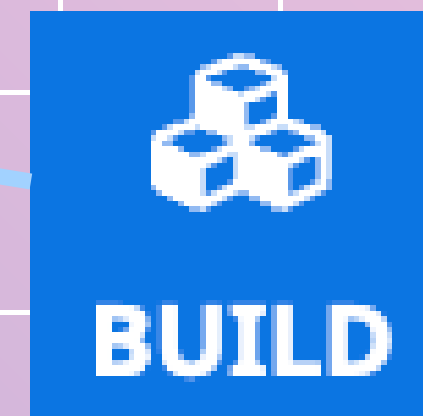
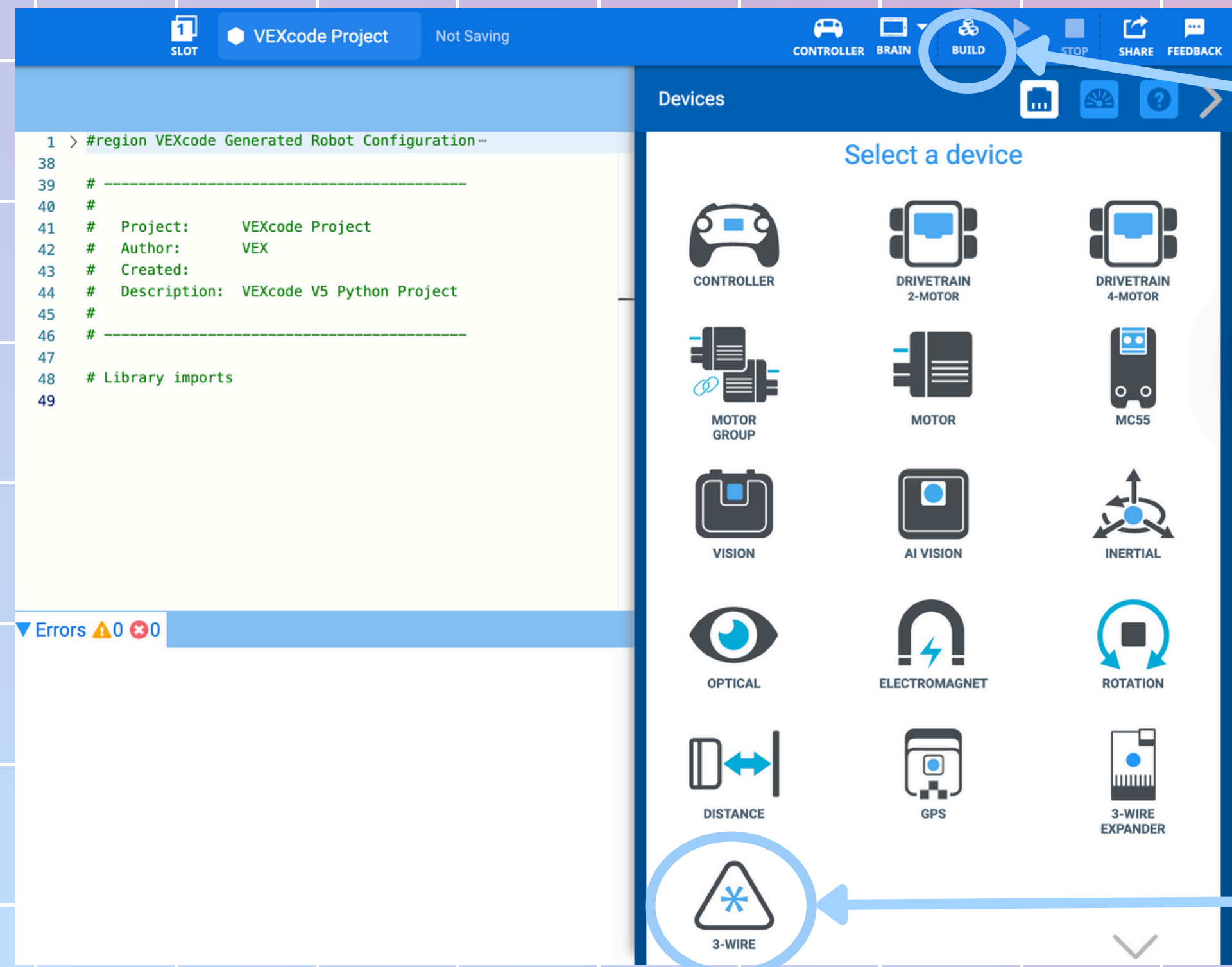
- A SENSOR IS A DEVICE THAT LETS A ROBOT GATHER INFORMATION ABOUT ITS ENVIRONMENT.
- SENSORS HELP ROBOTS MAKE DECISIONS INSTEAD OF JUST MOVING BLINDLY.

- WHY SENSORS MATTER

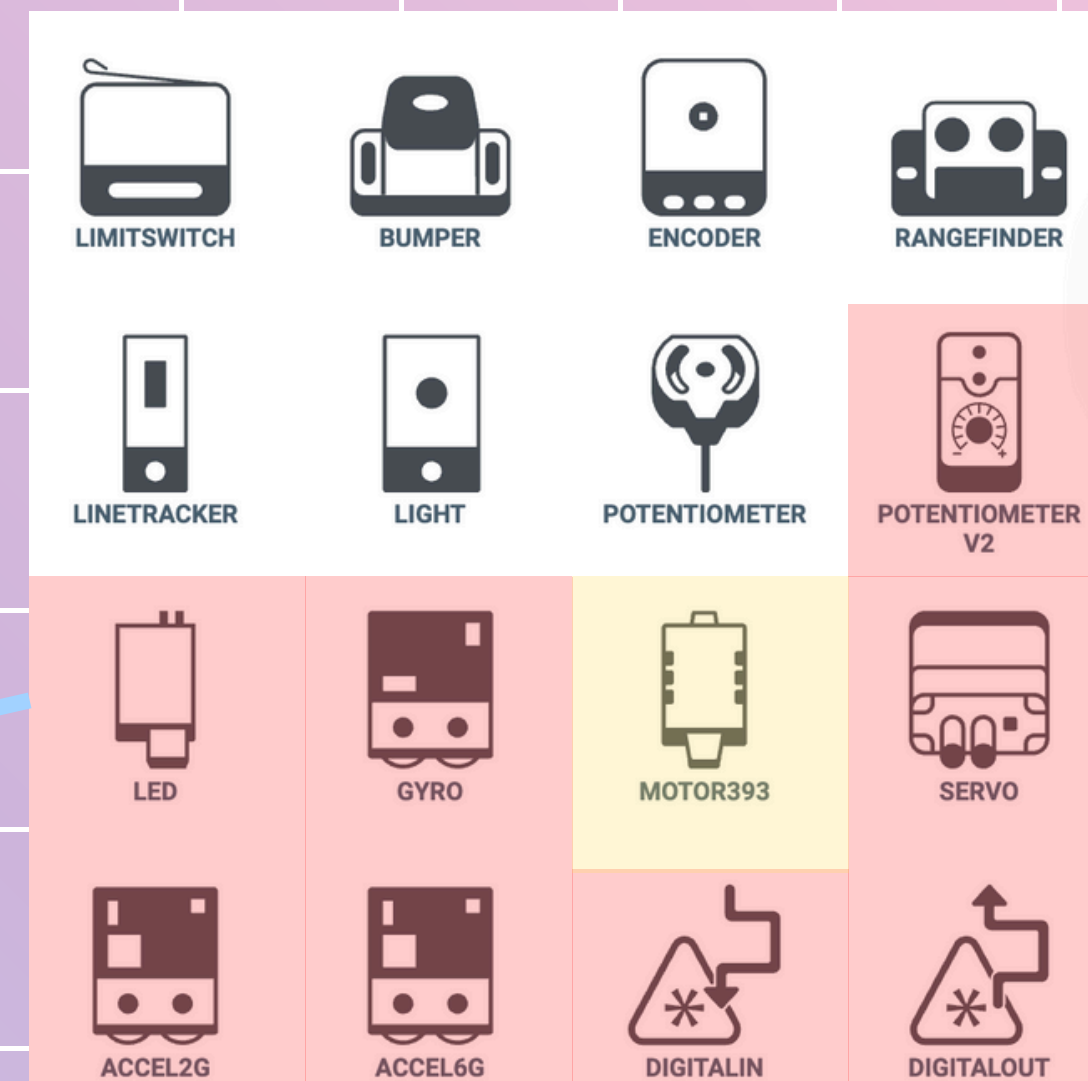
- THEY ALLOW AUTOMATION
- THEY MAKE ROBOTS MORE ACCURATE
- THEY REDUCE HUMAN CONTROL AND ERROR



VEX V5 Program



- COMPILES ALL OF THE
- DATA HELPS LOOK FOR ERRORS



Starting Programming With Sensors

Description

- BASIC PROGRAMMING IDEA
 - GET A SENSOR VALUE
 - COMPARE IT TO A CONDITION
 - TELL THE ROBOT WHAT TO DO
- EXAMPLE:
 - IF SOMETHING IS CLOSE → STOP
 - IF A BUTTON IS PRESSED → MOVE AN ARM
 - IF A COLOR IS DETECTED → TURN

**Most sensor
code follows
this structure:**

```
✓ if (sensor value meets condition) {  
    |   do something  
    }
```

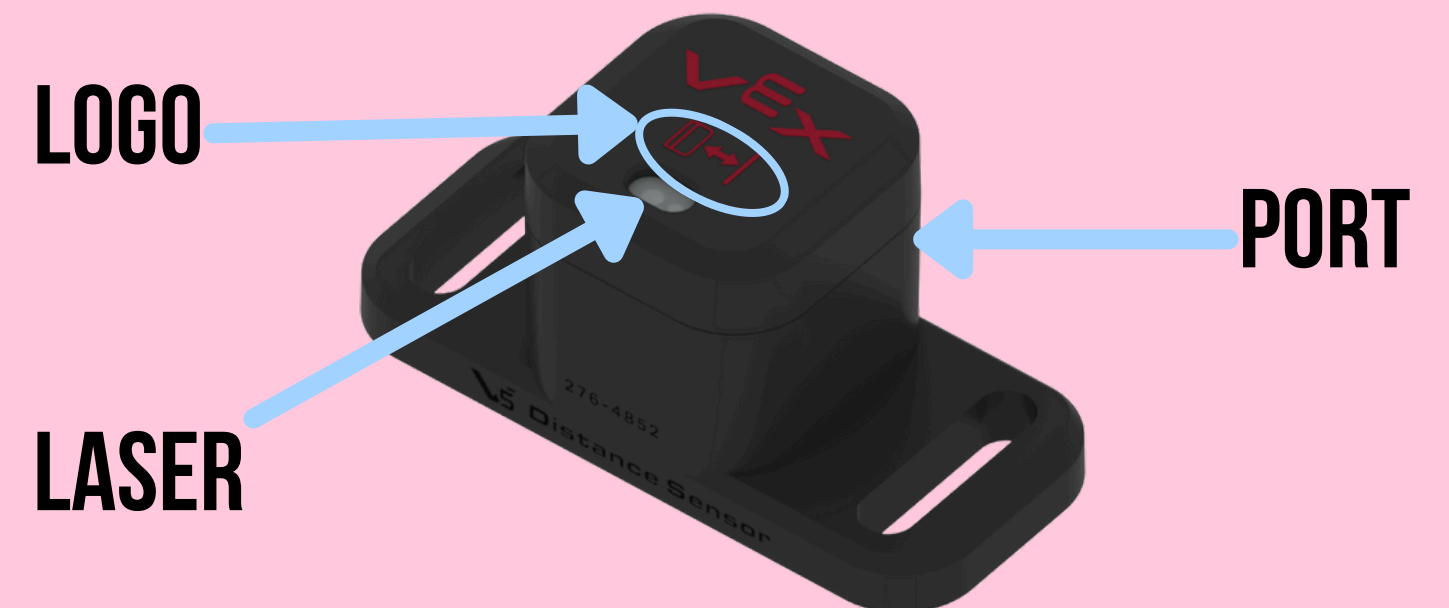
Distance Sensor

Description

- USES LASER TIME-OF-FLIGHT TECHNOLOGY TO MEASURE HOW FAR AWAY AN OBJECT IS.
- WHAT IT'S USED FOR
 - DETECTING OBJECTS
 - MEASURING DISTANCE TO OBJECTS
 - AUTOMATICALLY STOPPING OR STARTING AT CERTAIN DISTANCE

Example Code

```
✓ while distance_sensor.object_distance(INCHES) > 2.5:  
    left_motor.spin(FORWARD, 25, PERCENT)  
    right_motor.spin(FORWARD, 25, PERCENT)  
  
left_motor.stop()  
right_motor.stop()
```



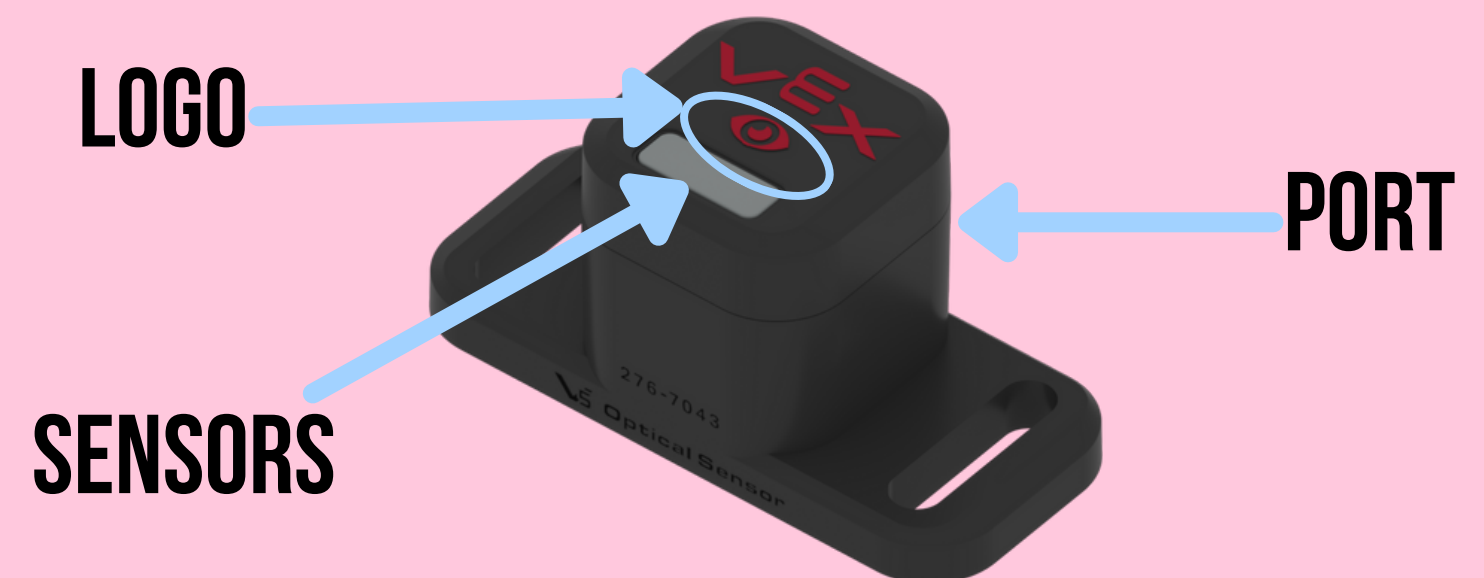
Optical Sensor

Description

- USES A COLOR SENSOR AND LIGHT SENSOR WITH A BUILT-IN LED.
- WHAT IT'S USED FOR:
 - IDENTIFYING COLORS
 - DETECTING OBJECTS PASSING IN FRONT
 - TRIGGERING ACTIONS BASED ON BRIGHTNESS

Example Code

```
✓ while True:  
✓     if optical.color() == Color.RED:  
✓         brain.screen.print("Red detected")  
✓     elif optical.color() == Color.BLUE:  
✓         brain.screen.print("Blue detected")  
        wait(200, MSEC)  
        brain.screen.clear_screen()
```



Limit Switch

Description

- A SMALL BUTTON THAT IS PRESSED WHEN SOMETHING TOUCHES IT.
- WHAT IT'S USED FOR:
 - DETECTING MECHANICAL LIMITS
 - KNOWING WHEN A MECHANISM HAS REACHED THE END
 - SIMPLE TOUCH DETECTION

Example Code*

```
limit_switch = Limit(brain.three_wire_port.b)
motor = Motor(Ports.PORT3)

while True:
    if limit_switch.pressing():
        motor.stop()
    else:
        motor.spin(FORWARD, 40, PERCENT)
```



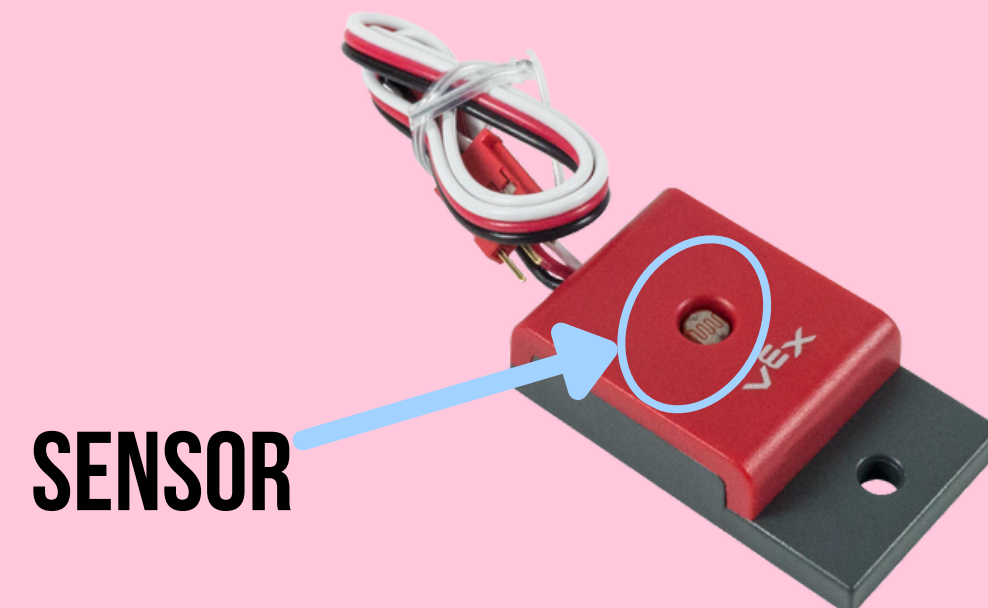
Light Sensor

Description

- MEASURES LIGHT INTENSITY IN THE ENVIRONMENT.
- WHAT IT'S USED FOR:
 - DETECTING BRIGHTNESS CHANGES
 - IDENTIFYING COLOR DIFFERENCES
 - LINE FOLLOWING

Example Code

```
while True:  
    brightness = light_sensor.brightness()  
  
    if brightness < 40:  
        brain.screen.print("Dark")  
    else:  
        brain.screen.print("Bright")  
  
    wait(200, MSEC)  
    brain.screen.clear_screen()
```



Shaft Encoder

Description

- MEASURES HOW MUCH A SHAFT HAS ROTATED.
- WHAT IT'S USED FOR:
 - TRACKING WHEEL ROTATION
 - MEASURING DISTANCE TRAVELED
 - MONITORING ARM OR JOINT POSITION

Example Code

```
✓ while True:  
    position = encoder.position(rotationUnits.DEG)  
    brain.screen.print("Rotation:", position)  
    wait(200, MSEC)  
    brain.screen.clear_screen()
```



Potentiometer

Description

- MEASURES ROTATIONAL POSITION USING VARIABLE RESISTANCE.
- WHAT IT'S USED FOR:
 - DETECTING EXACT ANGLES
 - CONTROLLING ARMS OR CLAWS PRECISELY
- ***DOES NOT TURN ALL THE WAY***

Example Code

```
brain = Brain()
pot = Pot(brain.three_wire_port.e)

while True:
    angle = pot.angle(rotationUnits.DEG)
    brain.screen.print("Angle:", angle)
    wait(200, MSEC)
    brain.screen.clear_screen()
```



Motor Controller 29

Description

- ADAPTER THAT ALLOWS V5 SYSTEMS TO CONTROL OLDER 3-WIRE MOTORS.
- WHAT IT'S USED FOR:
 - RUNNING LEGACY 393 MOTORS
 - EXPANDING MOTOR OPTIONS
- PROGRAMMING NOTE:
 - USED LIKE A NORMAL MOTOR ONCE CONNECTED THROUGH A 3-WIRE PORT



Ultrasonic Range Module

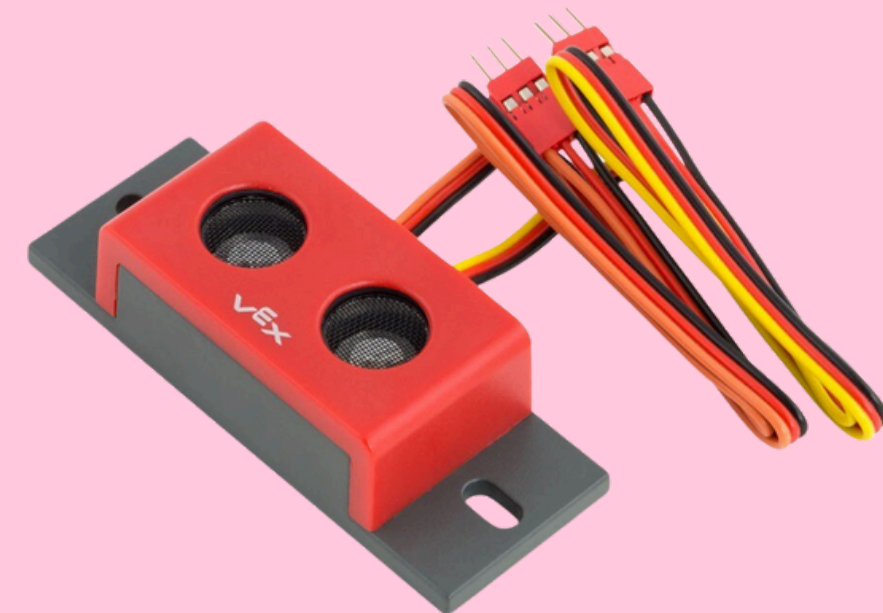
Description

- USES SOUND WAVES TO MEASURE DISTANCE (SIMILAR TO ECHOLOCATION).
- WHAT IT'S USED FOR:
 - MEASURING PROXIMITY
 - DETECTING OBJECTS
 - AVOIDING OBSTACLES

Example Code

```
brain = Brain()
sonar = Sonar(brain.three_wire_port.g)

while True:
    distance = sonar.distance(objectUnits.CM)
    brain.screen.print("Distance:", distance, "cm")
    wait(200, MSEC)
    brain.screen.clear_screen()
```



Line Tracker

Description

- DETECTS REFLECTED LIGHT FROM SURFACES.
- WHAT IT'S USED FOR:
 - LINE FOLLOWING
 - DETECTING LIGHT VS DARK SURFACES

Example Code

```
brain = Brain()
line_tracker = Line(brain.three_wire_port.h)

while True:
    reflect = line_tracker.reflectivity()
    if reflect < 40:
        brain.screen.print("Dark surface")
    else:
        brain.screen.print("Light surface")
    wait(200, MSEC)
    brain.screen.clear_screen()
```



Bumper Switch

Description

- A PHYSICAL BUTTON TRIGGERED WHEN PRESSED.
- WHAT IT'S USED FOR:
 - COLLISION DETECTION
 - END STOPS
 - SIMPLE TOUCH INPUT

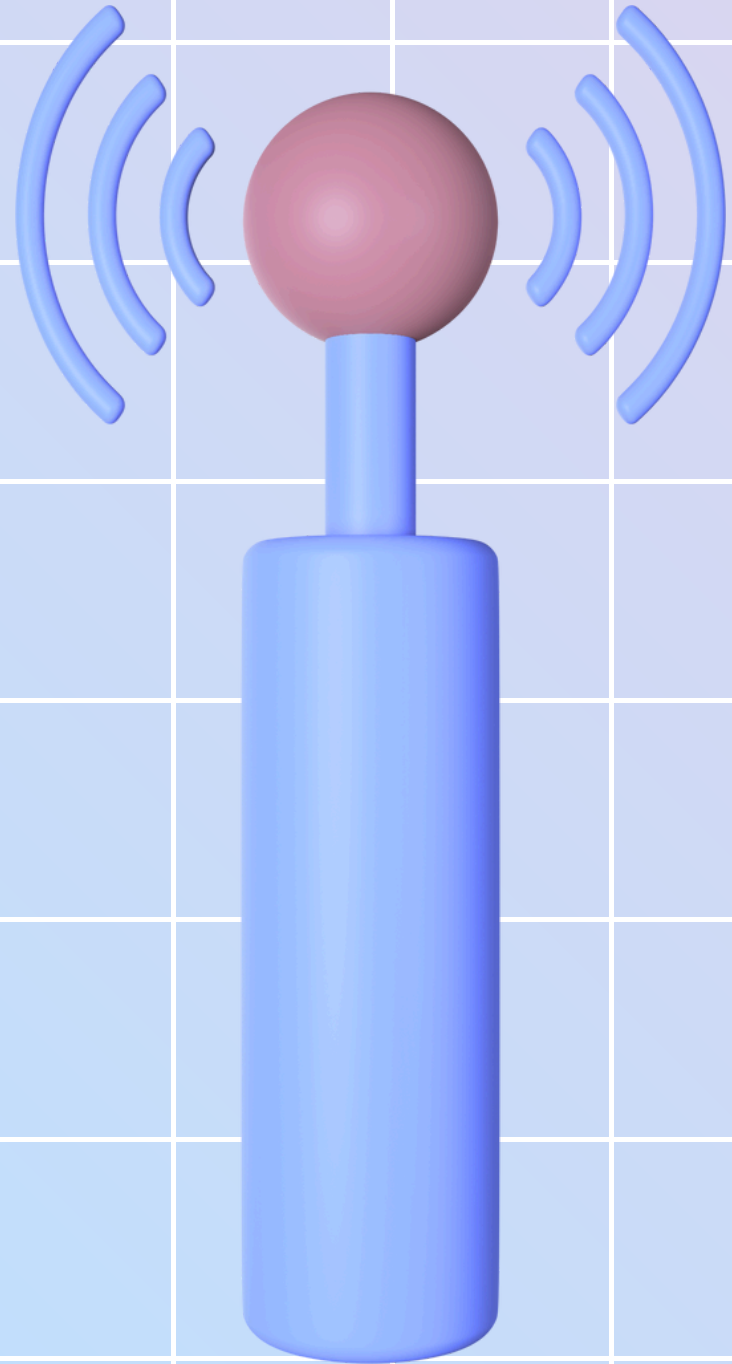
Example Code

```
brain = Brain()
bumper = Bumper(brain.three_wire_port.i)

while True:
    if bumper.pressing():
        brain.screen.print("Bumper pressed")
    else:
        brain.screen.print("Not pressed")
    wait(200, MSEC)
    brain.screen.clear_screen()
```



Using Sensors in Logic



- SENSORS ARE COMMONLY USED WITH:
 - IF / ELSE STATEMENTS
 - WHILE LOOPS
 - AUTONOMOUS BEHAVIORS
- EXAMPLE CONCEPT:
 - IF BUMPER IS PRESSED → STOP ROBOT
 - IF LINE TRACKER SEES BLACK → TURN



Key Takeaways

- SENSORS GIVE ROBOTS AWARENESS
- VALUES FROM SENSORS GUIDE DECISIONS
- VEX V5 PYTHON MAKES SENSOR PROGRAMMING READABLE AND POWERFUL
- SENSORS ARE ESSENTIAL FOR AUTOMATION AND CONSISTENCY

Questions?